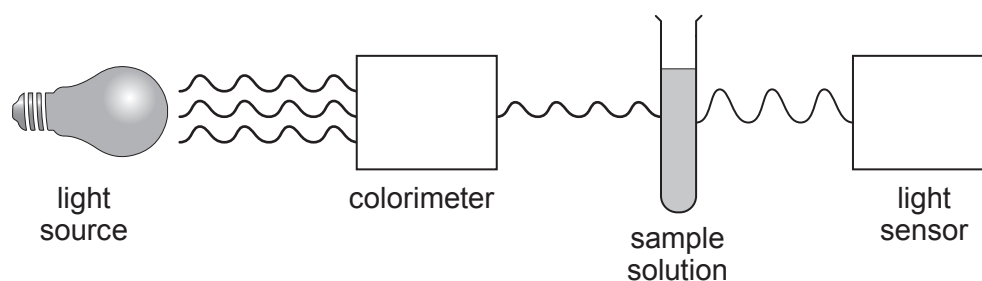


4. Lipases are involved in diverse biological processes, including the hydrolysis of triglycerides. Some lipases act in specific compartments inside cells, while others work in extracellular spaces.

(a) State the term given to enzymes which work within specific compartments inside cells.

[1]

Triacetin is a colourless triglyceride and can be hydrolysed by bacterial lipase. When an indicator is added to alkaline triacetin solution a coloured solution is produced. A colorimeter can then be used to monitor the progress of the reaction. When the triacetin is completely digested, the solution will change from coloured to colourless, allowing more light to pass through, and produce an absorbance reading of zero.



The effect of temperature on the hydrolysis of triacetin was investigated.

The experiment was performed as follows:

1. 2 cm^3 of triacetin was mixed with 3 cm^3 of sodium hydrogen carbonate (pH10) in a test tube and 5 drops of indicator added.
2. 1 cm^3 of lipase was placed in a separate test tube.
3. Both test tubes were placed in a water bath (20°C) for 5 minutes.
4. The contents of the tubes were combined, mixed thoroughly and 2 cm^3 transferred to a cuvette.
5. The time taken for the absorbance to reach zero was measured.
6. The experiment was repeated to give a total of three readings.
7. Steps 1-6 were repeated at 25 , 35 , 45 , and 50°C .

(b) (i) Identify the independent and dependent variables in this investigation.

[2]

independent

dependent



(ii) Explain why the solutions were left in the water bath for 5 minutes before combining and mixing. [1]

.....

.....

.....

(iii) Explain why it would be inappropriate to use a buffer solution in this experiment. [3]

.....

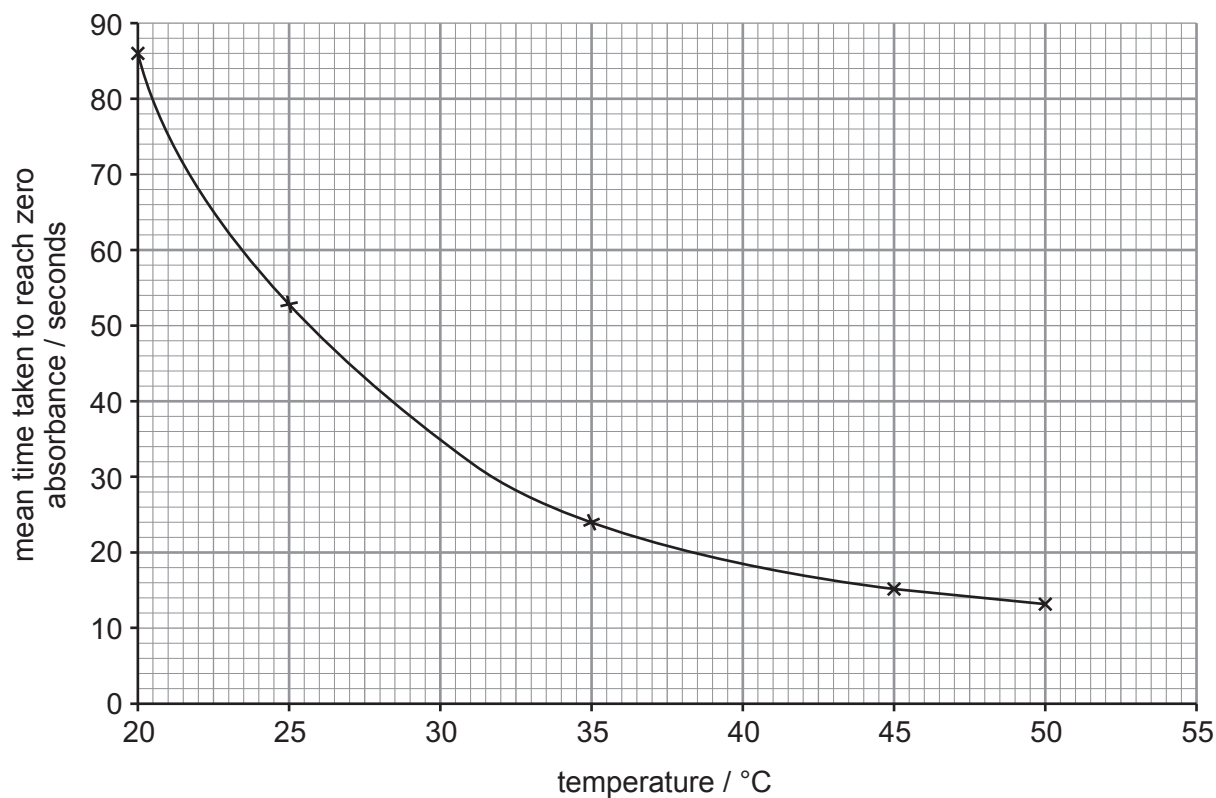
.....

.....

.....

.....

The results of the investigation are shown below.



Examiner
only

- (c) (i) In this experiment, 2 cm^3 of 0.25 g cm^{-3} triacetin was used. Calculate the rate of digestion per second at 40°C .

Give your answer in standard form. [3]

Rate = g s^{-1}

- (ii) Suggest **one** source of inaccuracy in this experimental method. [1]

.....

.....

.....

.....

- (iii) It was not possible to determine the optimum temperature for this enzyme in this experiment. Explain why and suggest how the design might be improved. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

15

